

SEONGKU KANG

Assistant Professor, Department of Computer Science and Engineering, Korea University

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RESEARCH INTERESTS

- Data science and machine learning for real-world applications
- Information retrieval and recommender systems

EXPERIENCE

- | | |
|---|-----------------------|
| Assistant Professor, Korea University <ul style="list-style-type: none">▪ Department of Computer Science and Engineering | Mar 2025 - Present |
| Postdoctoral Researcher, UIUC <ul style="list-style-type: none">▪ Data Mining Group (advisor: Prof. Jiawei Han) | Aug 2023 - Feb 2025 |
| Research intern, Microsoft Research Asia <ul style="list-style-type: none">▪ Social Computing group (mentors: Dr. Xing Xie, Dr. Jianxun Lian)· Award of excellence (Stars of Tomorrow, top 10%) | Oct 2022 - April 2023 |

EDUCATION

- | | |
|--|---------------------|
| Ph.D., Computer Science and Engineering, POSTECH <ul style="list-style-type: none">▪ Data Intelligence Lab (advisor: Prof. Hwanjo Yu) | Mar 2018 - Aug 2023 |
| B.S., Computer Science and Engineering, NTU <ul style="list-style-type: none">▪ Exchange Student | Jul 2016 - May 2017 |
| B.S., Computer Science and Engineering, Hanyang University <ul style="list-style-type: none">▪ Cumulative GPA: 4.43/4.50 (ranked 1st in CSE Dept.) | Mar 2012 - Feb 2018 |

PUBLICATIONS

Unless otherwise specified, the paper is accepted as a long/regular paper. * indicates equal contribution.

Conferences

- Topic Coverage-based Demonstration Retrieval for In-Context Learning
Wonbin Kweon, [SeongKu Kang](#), Runchu Tian, Pengcheng Jiang, Jiawei Han, Hwanjo Yu
EMNLP'25 | Conference on Empirical Methods in Natural Language Processing
- Scientific Paper Retrieval with LLM-Guided Semantic-Based Ranking
Yunyi Zhang, Ruozhen Yang, Siqi Jiao, [SeongKu Kang](#), Jiawei Han
EMNLP'25 Findings | Conference on Empirical Methods in Natural Language Processing
- DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning
Pengcheng Jiang, Jiacheng Lin, Lang Cao, Runchu Tian, [SeongKu Kang](#), Zifeng Wang, Jimeng Sun, Jiawei Han
COLM'25 | Conference on Language Modeling
- Imagine All The Relevance: Scenario-Profiled Indexing with Knowledge Expansion for Dense Retrieval
Sangam Lee, Ryang Heo, [SeongKu Kang](#), Dongha Lee
COLM'25 | Conference on Language Modeling

5. Delving into Instance-Dependent Label Noise in Graph Data: A Comprehensive Study and Benchmark
Suyeon Kim, SeongKu Kang[†], Dongwoo Kim, Jungseul Ok, Hwanjo Yu[†]
KDD'25 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
6. Embracing Plasticity: Balancing Stability and Plasticity in Continual Recommender Systems
Hyunsik Yoo, SeongKu Kang, Ruizhong Qiu, Charlie Xu, Fei Wang and Hanghang Tong
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval
7. Personalized Preference Reasoning with Large Language Models for Accurate and Explainable Recommendation
Jieyong Kim, Hyunseo Kim, Hyunjin Cho, SeongKu Kang, Buru Chang, Jinyoung Yeo and Dongha Lee
SIGIR'25 | ACM SIGIR Conference on Research and Development in Information Retrieval
8. Uncertainty Quantification and Decomposition for LLM-based Recommendation
Wonbin Kweon, Sanghwan Jang, SeongKu Kang[†], Hwanjo Yu[†]
WWW'25 | ACM The Web Conference
9. Chain-of-Factors Paper-Reviewer Matching
Yu Zhang, Yanzhen Shen, SeongKu Kang, Xiusi Chen, Bowen Jin, Jiawei Han
WWW'25 | ACM The Web Conference
10. Improving Scientific Document Retrieval with Concept Coverage-based Query Set Generation
SeongKu Kang, Bowen Jin, Wonbin Kweon, Yu Zhang, Dongha Lee, Jiawei Han, Hwanjo Yu
WSDM'25 | ACM International Conference on Web Search and Data Mining
11. Unsupervised Robust Cross-Lingual Entity Alignment via Neighbor Triple Matching with Entity and Relation Texts
Soojin Yoon, Sungho Ko, Tongyoung Kim, SeongKu Kang, Jinyoung Yeo, Dongha Lee
WSDM'25 | ACM International Conference on Web Search and Data Mining
12. Taxonomy-guided Semantic Indexing for Academic Paper Search
SeongKu Kang, Yunyi Zhang, Pengcheng Jiang, Dongha Lee, Jiawei Han, Hwanjo Yu
EMNLP'24 | Conference on Empirical Methods in Natural Language Processing
13. Continual Collaborative Distillation for Recommender System
Gyuseok Lee, SeongKu Kang, Wonbin Kweon, Hwanjo Yu
KDD'24 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
14. PEARL: A Review-driven Persona-Knowledge grounded Conversational Recommendation Dataset
Minjin Kim*, Minju Kim*, Hana Kim, Beong-woo Kwak, Soyeon Jeon, Hyunseo Kim, SeongKu Kang, Youngjae Yu, Jinyoung Yeo, Dongha Lee
ACL'24 Findings | Annual Meeting of the Association for Computational Linguistics
15. Self-Consistent Reasoning-based Aspect-Sentiment Quad Prediction with Extract-Then-Assign Strategy
Jieyong Kim*, Ryang Heo*, Yongsik Seo, SeongKu Kang, Jinyoung Yeo, Dongha Lee
ACL'24-short Findings | Annual Meeting of the Association for Computational Linguistics
16. Multi-Domain Sequential Recommendation via Domain Space Learning
Junyoung Hwang, Hyunjun Ju, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
SIGIR'24 | ACM SIGIR Conference on Research and Development in Information Retrieval
17. Learning Discriminative Dynamics with Label Corruption for Noisy Label Detection
Suyeon Kim, Dongha Lee, SeongKu Kang, Sukang Chae, Sanghwan Jang, Hwanjo Yu
CVPR'24 | IEEE / CVF Computer Vision and Pattern Recognition
18. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy
SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han

WWW'24 | ACM The Web Conference

19. Top-Personalized-K Recommendation
Wonbin Kweon, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
WWW'24 | ACM The Web Conference
20. Multi-Domain Recommendation to Attract Users via Domain Preference Modeling
Hyunjun Ju, SeongKu Kang, Dongha Lee, Junyoung Hwang, Sanghwan Jang, Hwanjo Yu
AAAI'24 | AAAI Conference on Artificial Intelligence
21. MvFS: Multi-view Feature Selection for Recommender System
Youngjune Lee, Yeongjong Jeong, Keunchan Park, SeongKu Kang[†]
CIKM'23-short | ACM International Conference on Information and Knowledge Management
22. Distillation from Heterogeneous Models for Top-K Recommendation
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
WWW'23 | ACM The Web Conference
23. Learning Topology-Specific Experts for Molecular Property Prediction
Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, Hwanjo Yu
AAAI'23 | AAAI Conference on Artificial Intelligence
24. Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering
SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu
WWW'22 | ACM The Web Conference
25. TaxoCom: Topic Taxonomy Completion with Hierarchical Discovery of Novel Topic Clusters
Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, Hwanjo Yu
WWW'22 | ACM The Web Conference
26. Obtaining Calibrated Probabilities with Personalized Ranking Models
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
AAAI'22 | AAAI Conference on Artificial Intelligence
27. Topology Distillation for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
KDD'21 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
28. Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, SeongKu Kang, Hyunjun Ju, Chanyoung Park, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
29. Unsupervised Proxy Selection for Session-based Recommender Systems
Junsu Cho, SeongKu Kang, Dongmin Hyun, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
30. Learning Heterogeneous Temporal Patterns of User Preference for Timely Recommendation
Junsu Cho, Dongmin Hyun, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference
31. Bidirectional Distillation for Top-K Recommender System
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference
32. DE-RRD: A Knowledge Distillation Framework for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
CIKM'20 | ACM International Conference on Information and Knowledge Management

33. Deep Rating Elicitation for New Users in Collaborative Filtering
 Wonbin Kweon, SeongKu Kang, Junyoung Hwang, Hwanjo Yu
 WWW'20-short | ACM International World-Wide Web Conference
34. Multi-Modal Component Embedding for Fake News Detection
SeongKu Kang, Junyoung Hwang, Hwanjo Yu
 IMCOM'20 | IEEE International Conf. Ubiquitous Information Management and Communication
35. Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users
SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu
 CIKM'19 | ACM International Conference on Information and Knowledge Management
 - ranked 13th among the most influential papers at CIKM 2019([link](#))
36. Densifying a Trust Network for Effective Collaborative Filtering
SeongKu Kang, Jemin Wang, Yeon-Chang Lee, Sang-Wook Kim
 KDBC'17 | Korean DataBase Conference

Journals

1. BPL: Bias-adaptive Preference Distillation Learning for Recommender System
SeongKu Kang, Jianxun Lian, Dongha Lee, Wonbin Kweon, Sanghwan Jang, Jaehyun Lee, Jindong Wang, Xing Xie, Hwanjo Yu
 IEEE Transactions on Knowledge and Data Engineering (SCIE)
2. Unbiased, Effective, and Efficient Distillation from Heterogeneous Models for Recommender Systems
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
 ACM Transactions on Recommender Systems (2024.02)
3. Mitigating Viewpoint Sensitivity of Self-Supervised One-Class Classifiers
 Hyunjun Ju, Dongha Lee, SeongKu Kang, Hwanjo Yu
 Information Sciences (SCIE) (2022.09)
4. Personalized Knowledge Distillation for Recommender System
SeongKu Kang, Dongha Lee, Wonbin Kweon, Hwanjo Yu
 Knowledge-Based Systems (SCIE) (2022.03)
5. Item-side Ranking Regularized Distillation for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
 Information Sciences (SCIE) (2021.11)

AWARDS & FELLOWSHIPS

Awards

- Outstanding reviewer, ACM KDD, 2024
- Stars of Tomorrow Award, Microsoft Research Asia, 2023
- POSTECH Research Award (4th place), National Research Foundation and POSTECH, Korea, 2022
- Music Playlist Recommendation Competition (3rd place), Kakao, 2020
- SIGIR Student Travel Award, ACM SIGIR, 2019-2020
- Dean's List, Hanyang University, 2018
- Best Paper Award (3rd place), Korean Database Conference (KDBC), 2017

Fellowships

- POSTECHIAN Fellowship, POSTECH, 2023
- NAVER Ph.D Fellowship, NAVER, 2021

- NAVER Academic Scholarship, NAVER, 2017
- National Science and Technology Scholarship, Ministry of Education, Korea, 2012-2017
- Hanyang Brain Scholarship, Hanyang University, 2012

PROFESSIONAL SERVICES

Program Committee/Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2023, 2024, 2025, 2026
- ACM International World-Wide Web Conference (WWW): 2024, 2025
- AAAI Conference on Artificial Intelligence (AAAI): 2024, 2025, 2026
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR): 2025
- Annual Meeting of the Association for Computational Linguistics (ACL): 2025
- SIAM Conference on Data Mining (SDM): 2024

Journal Reviewer

- Transactions on Information Systems: 2024
- Transactions on Knowledge and Data Engineering: 2024
- Knowledge-Based Systems: 2023, 2024
- Neural Computing and Applications: 2022

TALKS

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|---|----------|
| • Tutorial: Continual Recommender Systems, ACM CIKM | Nov 2025 |
| • Panel discussion, Data Intelligence Workshop | Jun 2025 |
| • CS/AI Seminar, Ulsan National Institute of Science and Technology (UNIST) | Nov 2023 |
| • Invited Seminar, Electronic & Information Research Information Center (EIRIC) | Sep 2023 |
| • Invited Seminar, Social Computing Group, Microsoft Research Asia (MSRA) | Jun 2023 |
| • Technology Showcase, Korea AI summit | Dec 2022 |